

TEACHER GUIDE

Teach the course companion

Sheet-metal Toolbox / Year 9 Industrial Technology — Metal

Coverage and purpose

The 50-page student book follows the existing five Toolbox modules and all 16 named theory sections. It retains all 20 original course multiple-choice questions, their option order and answers, all five written prompts and all 12 project folio prompts. Additional paper tasks teach comparison, plan interpretation, checkpoint reasoning and evidence-based evaluation. The answer material covers 83 student response IDs.

This is a unit companion. Relevant wider-course concepts assigned in the current Toolbox program are integrated without claiming to reproduce the entire 200-hour course. The whole-course hub's separate bank of 30 questions and three written tasks remain outside this unit book; the Toolbox's own questions are preserved completely.

Read, model, observe and apply

Explain the named section before asking students to apply it. Use the source plan, actual material samples and teacher demonstrations for information that a print illustration cannot establish. The worked examples on pages 6, 18, 30 and 36 deliberately separate hypothetical reasoning from the learner's own evidence. Open-task guidance describes credible evidence and acceptable alternatives, not one compulsory design.

Treat a precise unresolved question as useful evidence. Students may write "Teacher to confirm", "Not yet checked" or "Not yet observed" when that is accurate. Follow up with the necessary teaching or practical opportunity. Never turn a blank into an invented result or a paper response into automatic practical permission.

Print and edit

Print the controlled student PDF on A4, duplex on the long edge: 50 pages are 25 sheets. Preserve handwriting lines and drawing spaces. The native editable Word edition may reflow after changes; recheck pagination, source question wording and working space before printing. Screen inspection is not a physical print test.

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Confirm the controlling instructions

Resolve consequential details before the affected practical stage.

Drawing and project identity

The site name is Sheet-metal Toolbox. The authoritative drawing title is TOOL CARRYALL DEVELOPMENT. Page 12 contains a full-sheet source overview and enlarged complete drawing inset. Only surrounding page margins are omitted from the detail. All labels, views, dimensions, notes and title block remain source content. Students use written dimensions and current teacher instructions, never scale from a resized reproduction.

Confirm actual materials and stock, dimensions or line meanings that students cannot read, construction sequence, approved joining method, finish or lubricant, machine setup, safety procedures and permissions. The images do not prescribe mild steel, rivets or a coating. Tool photographs support recognition; they do not replace a demonstration or local procedure.

Timing and assessment boundaries

The website shows five two-week module labels, while the current v2.1 teaching program schedules 50 sixty-minute periods over 20 weeks. This book uses module sequence and leaves timetable decisions to the teacher. The repository also holds a 2026 assessment draft alongside newer program material. No draft marks, due dates or formal sign-off are imported into the book. Use the current teacher-issued task notification.

Properties studio

The program includes teacher-authorised heat-treatment or property-modification learning. Page 40 supplies an observation/reflection record, not an operating recipe. The teacher controls equipment, materials, method, SOP, permissions and any actual performance evidence. An honest pending observation remains pending; a written reflection cannot certify an unobserved practical skill.

Safety source roles

SafeWork NSW is the workplace regulator; WHS legislation and regulations establish legal duties and requirements; relevant standards supply technical benchmarks; codes give practical guidance for achieving required standards. Current teacher-approved local procedures govern the student's task and permission. These roles are explained on page 5 and applied through its response.

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Sources and workshop systems

Answers and expected evidence / student pages 3, 4, 5

Student p3 / p03-source

The dimension must be checked against the current teacher-issued Toolbox drawing and instructions; an online product cannot change the authorised geometry. Cultural material needs appropriate permission from the relevant authority and agreed conditions/attribution; online visibility is not permission. The teacher should guide any proposed use.

Student p4 / p04-control

Glasses protect against some eye hazards but do not secure the workpiece or prevent its movement. The approved workholding and other demonstrated controls must address movement at its source, alongside directed PPE.

Student p5 / p05-setup

Responses should identify task-relevant checks in all four categories, for example secured hair/clothing and correct PPE; demonstrated guard/stop setup; approved workholding; clear work zone. Accept other valid local checks. Unconfirmed equipment or permissions must remain explicitly awaiting teacher confirmation. Distinguish SafeWork NSW as the workplace regulator from the teacher/local process authorising the actual student, machine, setup and task.

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Incidents, controls and safety check

Answers and expected evidence / student pages 6, 7, 8

Student p6 / p06-report

A complete hypothetical report distinguishes observation from assumed cause; identifies a plausible workshop condition; states stop/make safe/alert teacher; and proposes review of a relevant control. Avoid blame, invented injury treatment or unsupported claims about who caused the event.

Student p7 / p07-scenarios

1. Work-zone access/nearby material has changed: stop as needed, keep clear and seek teacher direction to restore the approved zone. 2. Workholding/position has changed: stop, make safe and alert teacher; do not compensate by holding or forcing it. 3. Machine/task permission and demonstration may not apply: do not begin; seek task-specific teacher instruction. All answers should connect the action to changed risk.

Student p7 / p07-review

Examples: restore a physical separation/work zone; use the teacher-approved holding device; restrict machine access to the demonstrated task. The proposed control must target the hazard, and a check should establish that the approved condition is restored. Teacher confirms the actual setup.

Student p8 / Question 1

A. To decide how serious the risk is — Risk level depends on both the chance of harm and how serious the possible harm could be.

Student p8 / Question 2

B. Stop, make the area safe and seek teacher direction — Existing controls may no longer be effective when the work area, equipment, material or people nearby change.

Student p8 / m01-written

During drilling, an unsecured workpiece could spin. This is possible and could cause a serious hand injury. I would use the approved vice or clamp and machine guard, follow the teacher's setup and wear the directed PPE. If anything changes, I would stop and seek teacher direction. Accept equivalent source-grounded reasoning. Project-specific facts require the current teacher-issued instructions.

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Materials, brief and planning

Answers and expected evidence / student pages 9, 10, 11

Student p9 / p09-property

Example: ductility can allow a part to be formed without cracking; the actual material, thickness, approved bend and workshop process must still be confirmed. Other relevant properties are acceptable if connected to function and a legitimate information gap.

Student p10 / p10-brief

Entries must be drawn from the actual current brief. Requirements are mandatory functions/features; preferences are desirable choices; constraints limit resources/processes; evidence is an authorised way to judge a criterion. Do not accept an invented dimension or test as a confirmed requirement.

Student p11 / p11-plan

Read plan: orientation, labels, dimensions and questions confirmed; retain annotation. Prepare marking: approved datum, orientation, dimensions, waste side and permission; retain checked marking record. Before joining: approved method, trial-fit alignment/contact/position; retain trial-fit evidence. Exact sequence and approvals are teacher-controlled.

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Reading the original drawing

Answers and expected evidence / student pages 13, 14

Student p13 / p13-trace

The sketch should identify a real named component, link a written dimension to its actual controlled feature and distinguish outline/fold/edge information. Unknown line meanings should be flagged for teacher confirmation. Assess the information relationship rather than a guessed numerical value.

Student p14 / p14-find

Title must be TOOL CARRYALL DEVELOPMENT; orientation should make title/views readable. HANDLE and BODY must link to their actual labelled views. SIDE x 2 indicates two side pieces. Dimension relationship must follow the actual arrows/extension lines. Any ambiguous fold/edge interpretation remains a precise teacher question.

Student p14 / p14-question

Accept a precise source-based question about a dimension, fold/edge line, orientation, repeated part or view relationship. It should name the feature and connect the uncertainty to a possible marking/cutting/forming error. "Is it right?" alone is insufficient.

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Original plan and material checks

Answers and expected evidence / student pages 15, 16, 17

Student p15 / Question 3

A. A property is a metal characteristic; performance is how it behaves in use — Properties describe characteristics while performance describes how effectively the metal works in context.

Student p15 / Question 4

C. Its properties suit function, fabrication, corrosion, availability and finish — A sound justification connects relevant properties to the full project context.

Student p15 / Question 5

C. The drawing title and upright views — The title should read normally and the views should be upright before the drawing is interpreted.

Student p15 / Question 6

A. TOOL CARRYALL DEVELOPMENT — The verified title printed on the teacher-issued drawing is “TOOL CARRYALL DEVELOPMENT”.

Student p16 / Question 7

D. As different names for the matching project context — The website uses “Sheet-metal Toolbox”, while the drawing retains its printed title.

Student p16 / Question 8

B. HANDLE, BODY and SIDE x 2 — The verified component labels are HANDLE, BODY and SIDE x 2.

Student p16 / Question 9

B. Use the printed component label — Printed names and visible view information provide stronger evidence than assumptions about shape or size.

Student p16 / Question 10

A. From arrows and extension lines to the controlled feature — Dimension arrows and extension lines connect each value to the feature it controls.

Student p17 / Question 11

D. Fold or edge lines — Fold and edge lines carry different information from the component’s outside shape.

Student p17 / Question 12

C. Record the uncertainty for teacher checking — Unclear information should be recorded and checked rather than guessed or silently ignored.

Student p17 / Question 13

C. Exact dimensions, materials, settings, sequence and permissions — These project details and permissions must come from the authorised teacher sources.

Student p17 / Question 14

A. A checked plan annotation or folio note — The saved annotation or folio note shows that the plan was read and verified before marking out.

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Material justification and marking

Answers and expected evidence / student pages 18, 19, 20

Student p18 / m02-written

The teacher-issued plan specifies [metal] for [part]. Its [relevant property] suits the part's function, and it can be fabricated using the approved workshop process. I would also explain how the material suits the expected corrosion conditions, availability and finish. Accept equivalent source-grounded reasoning. Project-specific facts require the current teacher-issued instructions.

Student p18 / p18-gap

Accept a specific missing specification, observed material behaviour, reliable corrosion information, availability evidence or confirmed finish/process detail. The answer should explain why it affects the judgement.

Student p19 / p19-datum

An incorrect datum shifts related marks, so features may be incorrectly located or fail to align with adjoining parts. Check the approved datum, dimensions, orientation, finished boundary and waste side against the issued drawing; seek teacher confirmation where unclear.

Student p20 / p20-tools

Answers must pair actual authorised features with suitable demonstrated tools: for example length with rule, perpendicular reference with square, layout line with scribe or approved hole location with centre punch. Tool choice alone does not establish permission. Accept other teacher-approved pairings with a defensible function.

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Cutting, forming and checkpoints

Answers and expected evidence / student pages 21, 22, 23, 24

Student p21 / p21-order

Possible reasons: confirm outline/dimensions/orientation before shape changes; check and treat burrs by the approved process; verify the next fold and its access; ensure marks remain interpretable. Each reason should connect the checkpoint to later safety, accuracy or access.

Student p22 / p22-match

Steel rule: measure, graduated scale. Scriber: mark, hardened point making layout lines. Tin snips: cut, paired cutting blades. Manual nibbler: cut, cutting head removes small pieces. Bossing mallet: form, smooth rounded head. Accept classifications tied to a demonstrated legitimate use; no unsupervised technique should be inferred.

Student p22 / p22-check

Identification does not prove suitability. Confirm specified material/stock, feature, tool type/condition, approved setup, demonstration and current permission. Do not try extra force or an unapproved alternative.

Student p23 / Question 15

A. Provide one reliable reference for marking — A datum gives measurements a consistent starting point and reduces accumulated errors.

Student p23 / Question 16

A. Secure the work and recheck markings — Securing the work and checking datum, orientation, waste side and sequence support safe, accurate processing.

Student p23 / m03-written

I measure from one datum edge so each mark has the same reliable starting point. I mark the waste side to protect the finished size, follow the planned sequence so later features still line up, and secure the work so it cannot move while I cut or form it. Accept equivalent source-grounded reasoning. Project-specific facts require the current teacher-issued instructions.

Student p24 / p24-marking

Sketch must correspond to an approved actual feature, with correctly identified datum/reference, finished boundary and relevant waste. All numbers must come from the issued plan or current teacher instruction. A precise unresolved question is valid. Do not infer project geometry from the generic diagram on page 19.

Student p24 / p24-record

A valid record names a feature, controlling source and separate check of datum/orientation/markings/dimension; distinguishes a real observation from a plan; and identifies teacher confirmation where needed. Accept “Not yet checked” with a clear next action.

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Joining, machining and finishing

Answers and expected evidence / student pages 25, 26, 27, 28

Student p25 / p25-factors

Example: a permanent neat joint might conflict with a need for disassembly; evidence about access and maintenance would matter. Or heat-based joining might conflict with coating preservation; material/coating compatibility and teacher-approved method would matter. Accept any coherent factor pair, grounded in approved constraints.

Student p26 / p26-decisions

1. Permission/training/setup does not automatically transfer; seek teacher direction before use. 2. Stop immediately, make safe and alert teacher; approved workholding/position must be reviewed. 3. Do not start; seek demonstration and task-specific permission. Answers should explain the changed or missing control.

Student p27 / p27-finish

Expected reasoning connects purpose to material/process, corrosion or friction; identifies maintenance, preparation, cost, safety or environmental trade-off; and cites actual product information/current teacher instructions as needed. No specific coating/lubricant is prescribed.

Student p28 / p28-matrix

Questions should establish actual function/load criteria; assembly access and need for disassembly; material/coating compatibility, heat effects and preparation; worker controls, consumables, waste handling, maintenance or life-cycle impacts. These are information questions, not invented specifications or unauthorised process instructions.

Student p28 / p28-justify

Accept a specific question with two plausible implications. Example: if access prevents the approved joint from being formed, the teacher must review the sequence or approved method. Or coating compatibility may require a different approved preparation. Avoid treating speed alone as sufficient evidence.

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Joint checks and quality evidence

Answers and expected evidence / student pages 29, 30, 31

Student p29 / Question 17

A. Load, permanence, access, heat, appearance and corrosion — A suitable joint must match functional, manufacturing and environmental needs.

Student p29 / Question 18

A. Check fit and alignment before errors are locked in — A trial fit reveals contact, alignment and position problems while they can still be corrected.

Student p29 / m04-written

The joining method must carry the expected load and suit whether the part needs to come apart. I would also check access, heat effects, appearance and corrosion. Before permanent joining, I would trial fit the parts to check contact, alignment and position. Accept equivalent source-grounded reasoning. Project-specific facts require the current teacher-issued instructions.

Student p30 / p30-diagnose

Observation: corresponding edges do not align. Possible causes might include orientation, marking, cutting or forming variation. Evidence must discriminate, such as checking orientation against views and comparing relevant written dimensions/marks under teacher direction. Do not reward a guessed cause or unapproved correction.

Student p31 / p31-evidence

Alignment: annotated trial-fit observation/photo/sketch, but not an unperformed strength test. Correction: before/after records plus repeated approved check, but not proof of unrelated criteria. Dimension: actual measurement compared with written approved dimension, but not all functional/finish criteria. Accept equivalent evidence with honest limits.

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Technology and evidence chains

Answers and expected evidence / student pages 32, 33, 34

Student p32 / p32-forecast

Accept a logical, qualified chain, such as waste-reduction pressure → digital nesting → file verification/material-planning skills, with uncertain cost, availability, training or actual measured benefit. Avoid certainty that all jobs disappear or that every advanced method is automatically better.

Student p33 / p33-caption

A strong caption identifies the actual evidence/stage, a relevant feature or decision and its significance. It states a limit without inventing results. If no project evidence exists yet, a clearly labelled planned caption is acceptable for learning but is not production evidence.

Student p33 / p33-chain

Examples: requirements to final evaluation; working drawing to marking evidence; production plan to recorded change; quality check to problem solving. The explanation must show how one supplies criteria, information or evidence for the other.

Student p34 / p34-compare

One-off may favour flexibility, direct skill practice and limited setup; batch may justify validated digital preparation and repeatability. Both require appropriate accuracy, material, checking, trained supervision and resources. No fixed speed, cost or accuracy claims without evidence. People/skills should include setup, material judgement and process-specific training.

Student p34 / p34-decision

Example: conventional method may suit a single approved practice part when flexible setup and skill learning matter; it still needs demonstrated capability and checking. CNC might become more suitable if repeat quantity, suitable equipment and validated files justify setup. Accept opposite conclusions supported by clear contextual evidence.

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Technology check and evaluation

Answers and expected evidence / student pages 35, 36

Student p35 / Question 19

B. By suitability for the task and result — Suitability depends on accuracy, quantity, material, resources and purpose.

Student p35 / Question 20

C. The trial fit showed misalignment, so I rechecked it; next time I would check earlier — A strong evaluation names evidence, explains the result and identifies a realistic improvement.

Student p35 / m05-written

Hand cutting can suit a one-off part because it is flexible and develops practical skill. CNC can suit a batch when repeatability and accurate digital information matter. My [measurement, trial fit or photo] showed _____. Next time I would _____ so the result better meets the approved criteria. Accept equivalent source-grounded reasoning. Project-specific facts require the current teacher-issued instructions.

Student p36 / p36-repair

Example: “Against the edge-alignment criterion, my hypothetical trial-fit sketch shows one mismatch. This criterion is partly met, although appearance does not establish strength. I would seek teacher direction, recheck the related marks and repeat the approved trial fit before permanent joining.” Accept other criteria if explicitly approved, with labelled hypothetical evidence and no invented test result presented as real.

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Folio: brief and requirements

Answers and expected evidence / student pages 37, 38

Student p37 / folio-01

Evidence should name the current teacher-issued brief, intended user, need and function. It must distinguish confirmed requirements from assumptions. Accept concise original wording that accurately represents the brief; do not require an invented user story.

Student p37 / p37-evidence

Source should be identifiable teacher-issued document/instruction. Evidence must relate to function and use and be approved, not an invented load test or unsupported general claim.

Student p38 / folio-02

All entries must trace to the current brief or be clearly identified as student preferences/uncertainties. Requirements are mandatory; preferences are optional; constraints relate to resources, time, material or approved processes. Do not accept guessed dates, cost allowances or dimensions as confirmed facts.

Student p38 / p38-criteria

Each requirement should link to a suitable authorised check, with no invented criterion or test procedure. Answers may remain pending if teacher confirmation is needed; the record should identify exactly what is unresolved.

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Folio: risk and materials

Answers and expected evidence / student pages 39, 40

Student p39 / folio-03

The record identifies an actual task, relevant hazard, likelihood and consequence; chooses stronger controls before PPE; identifies task-directed PPE and permission/demonstration points; and includes review when conditions change. Teacher confirms local controls. A completed record does not authorise equipment use.

Student p39 / p39-risk

Rows should form one coherent risk case. Control choice should address the identified hazard rather than list generic PPE. Changed-condition row can be a real observation or clearly identified future review trigger. Local procedures and teacher confirmation remain required.

Student p40 / folio-04

Use actual specified material choices and credible observations/information. Link at least two relevant properties to function, fabrication, corrosion, availability or finish. A valid comparison acknowledges a trade-off and any unconfirmed specification; no compulsory alloy is inferred from illustration.

Student p40 / p40-material

Entries must identify material/source and explain a property-to-performance relationship. Observed appearance alone is not proof of alloy, strength or corrosion performance. "Not tested" with a suitable next evidence step is acceptable.

Student p40 / p40-studio

Require actual teacher-authorised performance/observation details, intended property change, specific observed evidence and a bounded interpretation. No heat recipe, temperature, timing, cooling medium or operating instructions are supplied here. If not observed, preserve that status. A paper reflection alone cannot certify practical achievement.

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Folio: research and drawing

Answers and expected evidence / student pages 41, 42

Student p41 / folio-05

Look for useful source references, annotations identifying what was learned, and concept evidence connected to the brief. Reasons for the approved direction should use requirements rather than taste alone. No copied cultural motif without appropriate permission; no research image may silently replace original plan geometry.

Student p41 / p41-direction

A convincing explanation links the direction to confirmed function, material, process or constraints. Rejected idea should be considered fairly and declined for a clear reason. Exact plan changes need teacher approval; valid alternatives in research do not imply authorisation.

Student p42 / folio-06

Look for labelled HANDLE, BODY and SIDE x 2; correct plan orientation; annotations connecting dimensions/extension lines to controlled features; distinction between outlines and fold/edge lines; clear recorded questions and teacher confirmations. Drawings must reference actual approved information. Do not accept invented measurements, scale measurements from the reproduction or unrecorded approvals.

Student p42 / p42-confirm

A valid record identifies the precise detail and actual confirmation/date, or honestly records that confirmation remains outstanding. This is evidence of consultation, not an automatic practical permission.

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Folio: resources and planning

Answers and expected evidence / student pages 43, 44

Student p43 / folio-07

HANDLE, BODY and SIDE x 2 should match the original drawing. Materials/dimensions must be transcribed accurately from current authorised information, with source identified. Tool/consumable entries need current teacher confirmation. Unknown specifications remain explicit; do not infer mild steel, rivets, coating or allowance from images.

Student p43 / p43-resource

A specific missing material, dimension interpretation, approved tool, consumable, allowance or process detail is valid. Explain its effect on safe setup, fit, production planning or finish. Do not fill the gap with a guess.

Student p44 / folio-08

Sequence must follow current teacher direction and be plausible for actual part relationships/access. Include pre-marking/cutting checks, approved forming checkpoints and trial fit before permanent joining where appropriate. Demonstrations/permissions are actual requirements, not invented sign-offs. A planned approval is not recorded as granted.

Student p44 / p44-change

Require a real dated change, reason/evidence and teacher confirmation as appropriate, or honest no-change status. Do not reward invented problems to fill the field.

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Folio: production and quality

Answers and expected evidence / student pages 45, 46

Student p45 / folio-09

Expect authentic evidence spanning applicable marking, cutting, forming, joining, machining and finishing stages, with captions explaining process decisions or checks. Identify any stage not undertaken or recorded; do not invent evidence. Refer to attached pages if more space is needed, using clear IDs and dates.

Student p45 / p45-index

Index entries should link to real accessible evidence, name the stage/date and explain its claim. A missing stage is marked not undertaken/not recorded with next action, rather than implied complete.

Student p46 / folio-10

Criteria must be current and teacher-approved; results must be actual measurements/observations/trial-fit evidence. Judgements should compare results to criteria and state limitations or next action. No invented tolerances, load tests or result values. "Not yet checked" is preferable to fabricated completion.

Student p46 / p46-limit

Accept an evidence gap such as appearance not establishing strength, an unmeasured feature or a missing repeat trial fit. The further check must be proposed for teacher approval, with no unsupervised testing procedure.

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Folio: problems and evaluation

Answers and expected evidence / student pages 47, 48

Student p47 / folio-11

A complete record identifies a real issue; specific supporting evidence; considered cause or alternatives; teacher-directed decision; actual approved correction; and safe test/check result. Causation must not be claimed without evidence. If unfinished, state the remaining action and permission rather than a fabricated success.

Student p47 / p47-before-after

Before/after evidence should refer to the same relevant feature or criterion and show an actual recorded change. It may state that the approved correction/check is pending. No new dimension or repair process should be invented.

Student p48 / folio-12

Evaluation should judge function, accuracy, appearance and construction quality against current approved criteria; cite specific genuine evidence; distinguish strengths, limitations and untested claims; and propose at least one realistic improvement with a reason. Accept mixed or incomplete outcomes when honestly evidenced; do not require a universally positive judgement.

Student p48 / p48-evaluation

Rows should point to real student evidence and show how it supports the stated judgement. A limitation identifies what the evidence cannot prove. Improvement should address a stated weakness and remain a proposal until tested.

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Use media as an observation task

All eight course-library references are indexed on student page 50.

Preview the selected clip and present it at the matching teaching point. Its role is conceptual support; the current workshop SOP, teacher demonstration and task permission remain controlling. Students can complete the book's teaching and paper reasoning without independently opening the website.

Module / video ID	Watch-for purpose
1 / XOkPcLD5Soo	The Hierarchy of Controls — Notice why stronger controls are selected before PPE and reviewed when conditions change.
2 / PaGJwOPg2kU	Understanding Metals — Connect strength, ductility, hardness and manufacturing behaviour to a reasoned material choice.
3 / bByYRFZYAIA	Marking Out (Metal) — Watch how datum edges, clear lines and pre-cut checks support accuracy.
3 / uHjEuSeo0fc	Hand Deburring Sheet Metal — Observe why edges and drilled holes require approved preparation before trial fit, joining or coating.
3 / NAaHh_V3rX0	Sheet Metal Bending Basics — Notice how bend order, allowance and spring-back affect planning. Do not infer project allowances or settings.
4 / 1G8lGECoe1U	How to Use POP Rivets — Identify rivet parts and grip range only as relevant to an approved riveted joint. The clip does not prescribe rivets.
4 / q0CAfXV-YdY	What Is Corrosion and How to Stop It — Connect surface condition, moisture and a continuous protective barrier. Confirm the actual project material and finish.
5 / lEyNWtM6MW4	CNC Machining — Compare programmed movement, setup and repeatability with conventional methods.

Optional oral or separate-paper response: name one observed feature, explain how it supports the lesson concept, and state one project detail that still requires teacher confirmation. Expected evidence is a specific accurate observation and a valid boundary; it is not a copied operating sequence. The eight source clips are references, not a claim that online availability is permanent.

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Source and image credits

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Core course: <https://stevencowell.github.io/Yr-9-Metal/toolbox/> . Frozen source repository: <https://github.com/stevencowell/Yr-9-Metal/tree/949f90eb6bc28bcdd010fb43b50f27ae956ee1c9> . Source snapshot date: 25 September 2026. Original Tool Carryall Development drawing and Toolbox teaching program v2.1 are retained in the build source archive.

SafeWork NSW official references: <https://www.safework.nsw.gov.au/about-us> ; <https://www.safework.nsw.gov.au/legal-obligations/legislation> ; <https://www.safework.nsw.gov.au/resource-library/codes-of-practice> . These establish source roles; teachers confirm local application and procedures.

Source tool photographs

Jenny callipers: Glenn McKechnie, "OddlegCalipers.jpg", CC BY-SA 3.0. Source: <https://commons.wikimedia.org/wiki/File:OddlegCalipers.jpg> . Scribers: Glenn McKechnie, "Scribers.jpg", CC BY-SA 3.0. Source: <https://commons.wikimedia.org/wiki/File:Scribers.jpg> . Licence: <https://creativecommons.org/licenses/by-sa/3.0/> .

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The photo plates place unchanged source photographs side by side, resized proportionally without cropping. The course bossing-mallet image is identified by its source as AI-generated; an exact original model is not asserted. New editorial images are AI-generated illustrations, not real student work or construction authority. The original plan is reproduced from the course source without redrawing its information.